The Chain Rule

Basic chain rule applications

1 Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

b $f(x) = \sin(3x^2)$

c $f(x) = e^{5x}$

d $f(x) = \ln(2x + 7)$

2 Differentiate $f(x) = \sqrt{x^2 + 9}$.

The chain rule with trig and exponential functions

3 Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

4 Differentiate $f(x) = e^{\sin x}$.

5 Differentiate $f(x) = \tan(e^x)$.

Nested (multiple) chain rule

6 Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.

7 Differentiate $f(x) = (\ln(3x))^4$.

Chain rule combined with product/quotient rules

8 Differentiate $f(x) = x^2 e^{3x}$.

9 Differentiate $f(x) = \frac{\sin(2x)}{x}$.

Chain rule with tables of values

10 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

Chain rule with inverse trig and logarithms

11 Differentiate $f(x) = \sin^{-1}(2x)$.

12 Differentiate $f(x) = \ln(\sin x)$.

13 Differentiate $f(x) = \tan^{-1}(x^2)$.

Finding tangent lines using the chain rule

14 Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.